

Digital Distraction: Statistical Data Analysis of Daily Screen Time and Academic Performance

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Introduction

The daily use of digital devices continues to rise, with teenagers spending on average 8.5 hours a day in front of screens for entertainment alone [3]. Studies suggest high screen time impacts the quality of sleep and well-being of children and adolescents [2]. Since students use electronic devices not only for leisure but also for studying, their actual total daily screen times are likely higher. Considering these trends, this report investigates the research question: Is total daily screen time associated with the academic performance of students?

Data sample

The data analysis was performed on the 'Gaming vs Academic Performance' dataset found on Kaggle.com [1]. The dataset contains information on 8000 students aged 16-25 and their study and leisure time habits. The following columns were chosen for this analysis:

- `device_usage`: Total daily screen time including leisure and other device usage (in hours).
- `grades`: Academic performance score of the student on a scale from 0 to 100

Results and discussion

After basic data cleaning eliminating outlier grades above 100, the overall distribution of academic scores relative to daily screen times of the students was represented on a scatter plot as seen in Figure 1.

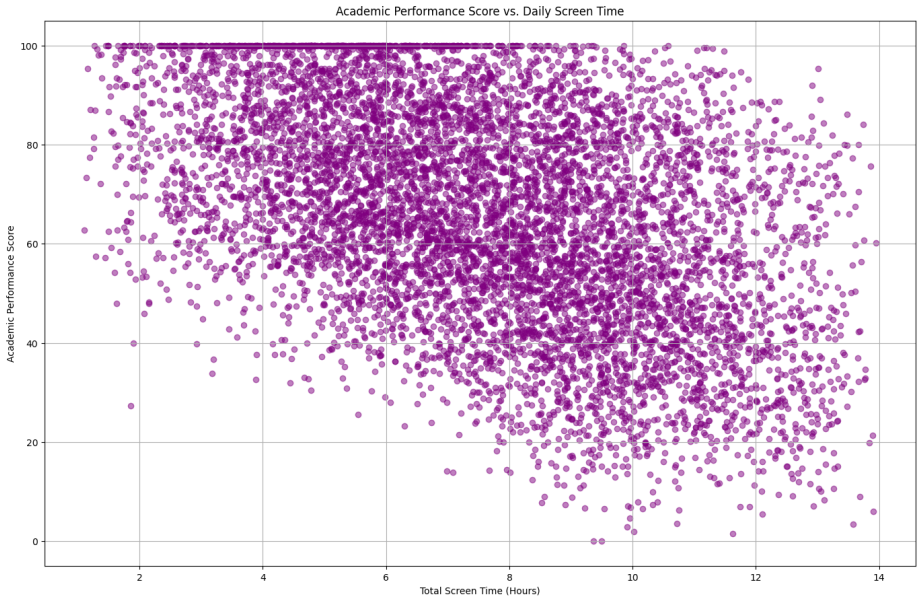


Figure 1: Academic Performance vs. Daily Screen Time

The frequency of scores below 40 rises dramatically among students with higher screen times. The clear negative trend in the data suggests that higher screen time is associated with lower academic performance. To quantitatively analyse this relationship, a Pearson's correlation coefficient was calculated :

$$r = -0.47, \quad p < 0.001$$

The results indicated a significant negative correlation between total daily device usage and academic performance of students. To further investigate this relationship, a median screen time was calculated: 7.6 hours. Based on it, two groups were created: low screen time (below median) and high screen time (above median). Descriptive statistics were used to calculate the average academic scores and statistical deviation for each group.

Group	**Average Grade**	**Standard Deviation**
Low Screen Time	74.75	18.07
High Screen Time	56.22	21.67

Table 1: Average academic scores and uncertainty for each screen time group.

As observed in Table 1 the lower screen time group achieved better academic scores on average than the high screen time group. The relationship between these two groups with their respective standard deviations can be seen in Figure 2.

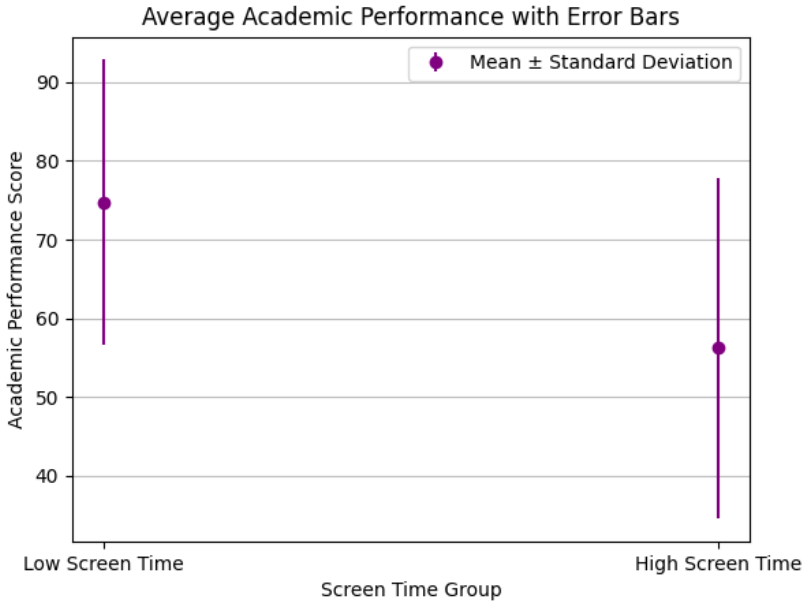


Figure 2: Average Academic Performance by Screen Time Group

While the error bars in Figure 2 represent a wide distribution of scores in both groups, the high screen time group displays a bigger variance in scores ($SD_{low} = 18.07, SD_{high} = 21.67$). This suggests that scores in the high screen time group become more unpredictable. However a clear negative trend can still be observed. To verify if these results are significant a t-test was performed:

$$t = 41.20, \quad p < 0.001$$

The analysis revealed a highly significant difference in academic performance between the low and high screen time groups. This further confirms the negative trend observed in the descriptive statistics and scatter plot are valid, supporting the conclusion that there is a statistically significant correlation between high screen time and lower academic performance.

References

[1] Gaming vs Academic Performance. (2026, June 26). https://www.kaggle.com/datasets/aiexplorer77/gaming-vs-academic-performance/data

[2] Hale, L., Guan, S. (2014). Screen time and sleep among school-aged children and adolescents: A systematic literature review. Sleep Medicine Reviews, 21, 50–58. https://doi.org/10.1016/j.smrv.2014.07.007

[3] Rideout, V., Peebles, A., Mann, S., Robb, M. B. (2022). Common Sense census: Media use by tweens and teens, 2021. San Francisco, CA: Common Sense